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Ivar Cashin Eriksson

Dear Normal Computing team,

I am applying for the AI Research Residency at Normal Computing. After several years working as a data scientist and ML engineer, I want to move back towards research and build a career investigating modern machine-learning systems. What I have enjoyed most throughout my work is taking poorly understood problems, developing ways to test them, investigating why models fail, and discussing difficult technical questions with colleagues. The opportunity to do this on agentic systems while contributing to publishable research is particularly appealing to me.

My background is in Engineering Physics, with a specialisation in Statistical Learning and Data Analytics. At RaySearch Laboratories, I researched machine-learning methods for automated radiotherapy planning using representation learning, Gaussian processes, and numerical optimisation. I developed latent patient representations, probabilistic prediction models, and optimisation methods, working closely with medical physicists and clinicians to evaluate their behaviour. This gave me experience taking open-ended research questions and turning them into experiments where both the performance and limitations of a method mattered.

Since then, I have worked primarily on applied machine learning and ML engineering. At Signum Life Science, I lead the development of ML systems spanning data validation, training, experiment tracking, evaluation, and batch inference, including a forecasting platform supporting thousands of independently trained models. I also develop embedding-based retrieval methods for domain-specific text, where much of the work involves designing evaluations and analysing failure cases. Previously at Valcon, I led ML projects from problem formulation through deployment and developed neural-network systems using transfer learning. Across these roles, I have become comfortable moving between research questions, experimental code, and maintainable systems.

Agentic systems and reinforcement learning are areas I particularly want to develop further. I am interested in how agents learn to use tools, how sequential decisions should be evaluated, and how feedback can be turned into useful learning signals rather than simply optimising convenient proxies. Code generation is especially interesting in this respect because success can often be tested directly: code can be executed, tools can return structured feedback, and failures can be analysed rather than judged only through subjective evaluation. I would be excited to investigate these questions experimentally while building agents and evaluation frameworks that are useful beyond a single research prototype.

Normal's residency appeals to me because it combines the parts of research and engineering that I most enjoy. The expectation that residents formulate their own technical questions, build reproducible implementations, evaluate them rigorously, and ultimately communicate the results through a paper is very close to how I would like to work. I would bring a strong mathematical foundation, substantial experience building ML systems, and a willingness to investigate ideas carefully rather than stopping when a demonstration appears to work. I would be excited to develop that experience further alongside researchers and engineers working at the frontier of agentic AI.

Kind regards,
Ivar Cashin Eriksson